

Multiple polylogarithms at Gaussian and Eisenstein arguments: closed forms, double-shuffle dimensions, and the onset of depth three

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Abstract

We study the double and triple polylogarithms $\text{Li}_{a,b}(z_1, z_2)$ and $\text{Li}_{a,b,c}(z_1, z_2, z_3)$ evaluated at fourth and third roots of unity—the Gaussian (level 4, $\mathbb{Q}(i)$) and Eisenstein (level 3, $\mathbb{Q}(\sqrt{-3})$) cases—using the multivariate special functions introduced in Wolfram Language 15 together with a high-precision experimental-mathematics pipeline. Three kinds of result emerge. First, the *stuffle* product gives a complete family of exact closed forms for the symmetric and diagonal combinations of the Gaussian and Eisenstein doubles, reducing them to single L -values; at the Eisenstein mixed point the colored tail degenerates to an *ordinary* zeta value. Second, the full *double shuffle*—stuffle together with the shuffle of Goncharov iterated integrals—collapses the depth-2 space far below what an integer-relation search alone can see: the Gaussian double points contribute only 5 new constants through weight 6 (not the 30 a precision-limited PSLQ reports), and the Eisenstein (ρ, ρ) family collapses to a *single* new constant. The discrepancy is a sharp methodological lesson: the genuine reductions are of large height and are therefore invisible to integer-relation search once the basis is large, whereas the derived double shuffle settles them exactly. Third, turning to depth three, we show that 7 of the 10 weight-6 Gaussian triples are depth- ≤ 2 products, and we identify the remaining 3 as genuinely depth-three by matching against the motivic dimension: Deligne’s theorem and Glanois’s depth-graded basis give, for both level 3 and level 4, a free motivic Lie algebra with one generator per weight, hence $d_n = 2^n$ and $\dim \text{gr}_p^D = \binom{n}{p}$, so that $\dim \text{gr}_3^D(\text{weight } 6) = \binom{6}{3} = 20 > 0$. Genuine depth three thus appears already at weight 6 in the Gaussian and Eisenstein towers—five weights below the classical threshold of the ordinary multiple zeta values. Every displayed identity has been verified to several hundred digits in two independent engines.

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1 Introduction

The classical polylogarithm $\text{Li}_s(z) = \sum_{n \geq 1} z^n / n^s$ has, since Lewin, a vast catalogue of functional equations and special values. Its multivariate generalization, the *multiple polylogarithm*

$$\text{Li}_{s_1, \dots, s_k}(z_1, \dots, z_k) = \sum_{n_1 > n_2 > \dots > n_k \geq 1} \prod_{i=1}^k \frac{z_i^{n_i}}{n_i^{s_i}}, \quad (1)$$

of *weight* $w = s_1 + \dots + s_k$ and *depth* k , has become a central object of analytic number theory and quantum field theory. When the arguments z_i are roots of unity, the values (1) are the *cyclotomic* (or colored) multiple zeta values; the case of all $z_i = 1$ recovers the multiple zeta values $\zeta(s_1, \dots, s_k)$, and the cases $z_i \in \mu_N$ (the N -th roots of unity) give the level- N values. This article concerns the two smallest non-trivial levels beyond the alternating ($N = 2$) case: the *Gaussian* level $N = 4$, where the relevant field is $\mathbb{Q}(i)$, and the *Eisenstein* level $N = 3$, with field $\mathbb{Q}(\sqrt{-3})$.

A practical novelty makes a systematic experimental study possible. Wolfram Language 15.0 (June 2026) provides first-class symbols for these functions—`MultiplePolyLog`, `MultipleZeta`, `GeneralizedPolyLog` (the Goncharov iterated integral), `HarmonicPolyLog`, `MultipleHarmonicNumber`—so that a colored double or triple polylogarithm can be typed, evaluated to arbitrary precision, and (where the kernel can) reduced symbolically. We pair this with a custom pipeline: accelerated evaluators for the regimes where the built-in functions time out, an arbitrary-precision integer-relation search, and—decisively for what follows—an implementation of the *double shuffle* that works with derived relations and exact rational rank rather than numerical relation-finding.

The three threads of the article are as follows.

Closed forms (Sections 2–3). The nested-sum (*shuffle*) product gives exact, elementary closed forms for the symmetric and diagonal combinations of the Gaussian doubles, reducing them to products of single polylogarithms at $i, -1, -i$. The Eisenstein analogue is identical in form, with one arithmetic surprise at the mixed cube-root point.

Double-shuffle dimensions (Sections 4–6). The *shuffle* of Goncharov iterated integrals supplies a second, independent family of relations. Imposing both—the double shuffle—collapses the colored depth-2 space dramatically. The Gaussian double points contribute exactly 5 genuinely new constants through weight 6; the Eisenstein (ρ, ρ) family contributes exactly 1. A precision-limited integer-relation search, by contrast, reports a far larger number, because the true reductions have large

height. We make this methodological point precise: *for high-height reductions the derived double shuffle is reliable where integer-relation search is not.*

The onset of depth three (Section 5). The depth filtration is where the arithmetic becomes most interesting. For the uncoloured multiple zeta values, genuine depth-three content first appears only at weight 11. We show that in the Gaussian and Eisenstein towers it appears already at weight 6. The computation has two halves: a shuffle calculation reducing 7 of the 10 weight-6 Gaussian triples to depth ≤ 2 , and an appeal to the motivic dimension (Deligne; Glanois) certifying that the remaining 3 are genuine.

Throughout, “verified” means the displayed identity was confirmed numerically to the stated precision in `mpmath` and, independently, in Wolfram 15.

2 Two products and the double shuffle

A multiple polylogarithm has two faces, and each carries a multiplication.

2.1 The stuffle product

Multiplying two nested sums (1) and re-sorting the summation indices by their relative order gives the *stuffle* (quasi-shuffle, or harmonic) product. In depth one it is the splitting of the lattice $\{(m, n)\}$ into $m > n$, $m < n$, $m = n$:

$$\mathrm{Li}_a(u) \mathrm{Li}_b(v) = \mathrm{Li}_{a,b}(u, v) + \mathrm{Li}_{b,a}(v, u) + \mathrm{Li}_{a+b}(uv). \quad (2)$$

The collision term $\mathrm{Li}_{a+b}(uv)$ multiplies the arguments—the feature that will move colored values between levels. Identity (2) is exact and holds wherever the three series converge.

2.2 The shuffle product

The second face is Chen’s iterated integral. With the Goncharov convention

$$G(a_1, \dots, a_n; 1) = \int_{0 < t_1 < \dots < t_n < 1} \prod_{i=1}^n \frac{dt_i}{t_i - a_i}, \quad (3)$$

the single polylogarithm is $\mathrm{Li}_n(z) = -G(0^{n-1}, z^{-1}; 1)$, and the depth-two polylogarithm is

$$\mathrm{Li}_{a,b}(z_1, z_2) = G\left(0^{a-1}, \frac{1}{z_1}, 0^{b-1}, \frac{1}{z_1 z_2}; 1\right), \quad (4)$$

with the analogous word $0^{a-1} y_1 0^{b-1} y_2 0^{c-1} y_3$, $y_j = (z_1 \cdots z_j)^{-1}$, in depth three. The product of two iterated integrals over the common interval $[0, 1]$ is the integral over the product region, which decomposes into the shuffles of the two letter-words:

$$G(\mathbf{u}; 1) G(\mathbf{v}; 1) = \sum_{\mathbf{w} \in \mathbf{u} \amalg \mathbf{v}} G(\mathbf{w}; 1). \quad (5)$$

Re-identifying each shuffle word through (4) expresses a product of polylogarithms as a $\mathbb{Q}$ -linear combination of polylogarithms—generally at *different* colored points from those the stuffle produces. Equating the two expansions of the same product yields the *double-shuffle relations*.

2.3 Method: derived relations versus integer-relation search

Both products are exact and have small integer coefficients. Given numerical values of the relevant polylogarithms (to several hundred digits), we generate every stuffle and shuffle relation among the products $\text{Li}_p(s) \text{Li}_q(t)$ of a fixed weight, verify each numerically to $\sim 10^{-600}$, and then compute the *exact rational rank* of the resulting linear system on the colored doubles, modulo the single polylogarithms and a chosen generating set. The dimension of the space is the corank.

This should be contrasted with the more usual approach of feeding each colored value to an integer-relation search (PSLQ or LLL) against a basis of candidate constants. That approach is indispensable for *discovering* an unknown closed form, but it has a precision ceiling: a genuine relation of height H is invisible to a search whose basis pushes the “spurious scale” $10^{\text{dps}/\#\text{atoms}}$ below H . As the number of basis atoms grows, the ceiling drops. We will see in Section 4 that the true reductions of the colored doubles are of large height, so that integer-relation search systematically *over*-counts their independence, while the derived double shuffle—working with the small-height generators and exact rank—returns the correct, much smaller, dimension. The lesson recurs throughout: *derived relations plus exact rank are reliable where high-precision relation-finding is not.*

3 Gaussian doubles: closed forms

Write $\text{Li}_{a,b}$ at the three Gaussian base points beyond $(i, 1)$:

$$(i, i), \quad (-1, i), \quad (i, -1),$$

all generated to 700 digits by an accelerated evaluator (Section 7) and agreeing with Wolfram 15 to full precision. Under the level-4 complex conjugation $i \mapsto -i$ the imaginary parts are odd and the real parts even.

3.1 Family 1: the stuffle closed forms

Specializing (2) at the Gaussian points ($uv = -1$ at (i, i) ; $uv = -i$ at the mixed pair) gives a complete set of exact closed forms for the symmetric combinations:

$$\text{Li}_{a,b}(i, i) + \text{Li}_{b,a}(i, i) = \text{Li}_a(i) \text{Li}_b(i) - \text{Li}_{a+b}(-1), \quad (6)$$

$$\text{Li}_{a,b}(-1, i) + \text{Li}_{b,a}(i, -1) = \text{Li}_a(-1) \text{Li}_b(i) - \text{Li}_{a+b}(-i), \quad (7)$$

$$\text{Li}_{a,b}(i, -1) + \text{Li}_{b,a}(-1, i) = \text{Li}_a(i) \text{Li}_b(-1) - \text{Li}_{a+b}(-i),$$

with the clean diagonal corollary

$$\text{Li}_{a,a}(i, i) = \frac{1}{2}(\text{Li}_a(i)^2 - \text{Li}_{2a}(-1)). \quad (8)$$

Every single polylogarithm on the right is elementary in the level-4 vocabulary: $\text{Im Li}_n(i) = \beta(n)$ is the Dirichlet beta value, $\text{Re Li}_n(i)$ and $\text{Li}_n(-1) = -\eta(n)$ are rational multiples of $\zeta(n)$ (or π^n at even n), and $\text{Li}_n(-i) = \overline{\text{Li}_n(i)}$. So Family 1 reduces every symmetric and diagonal Gaussian double to single L -values in closed form.

Remark 1. The three stuffle identities (6)–(7) were checked against the stored 700-digit values to $\sim 10^{-692}$ for every admissible a, b , and the closed forms were confirmed symbolically and numerically by Wolfram 15 at low weight, including the weight-6 diagonal $\text{Li}_{3,3}(i, i) = \frac{1}{2}(\text{Li}_3(i)^2 - \text{Li}_6(-1))$, for which the kernel returns a form in $\beta(4)$, π^4 , $\zeta(3)$.

The consequence is structural: all depth-2 content of the Gaussian doubles lives in the *antisymmetric* parts $\text{Li}_{a,b}(P) - \text{Li}_{b,a}(P)$, which vanish on the diagonal. These are what the double shuffle of the next section reduces.

3.2 Family 2: antisymmetric reductions to the $(i, 1)$ algebra

Against the basis of single-polylog monomials together with the $(i, 1)$ generators $g_{a,b} = \text{Im Li}_{a,b}(i, 1)$, $h_{a,b} = \text{Re Li}_{a,b}(i, 1)$, the *low-weight* antisymmetric parts reduce, each reduction a new closed-form identity found by integer-relation search and confirmed in Wolfram 15. A representative instance, with $\mu_n = \text{Re Li}_n(\frac{1+i}{2})$ and $\lambda_n = \text{Im Li}_n(\frac{1+i}{2})$:

$$\text{Re}[\text{Li}_{1,2}(i, i) - \text{Li}_{2,1}(i, i)] = 3\zeta(3) - 4\log 2 \mu_2 - \pi \lambda_2 - 4h_{1,1} \log 2 + 3h_{1,2}. \quad (9)$$

Such reductions exist at weights 2–4 at the mixed points and at weight 3 at (i, i) ; at higher weight the antisymmetric parts do not reduce *against the $(i, 1)$ basis alone*. To go further—and to learn how many are genuinely new—one needs the shuffle.

4 Gaussian doubles: the double-shuffle dimension

4.1 A cautionary integer-relation count

A first attempt to count the genuinely-new depth-2 directions proceeds by integer-relation search: test each antisymmetric part for membership in the span of the $(i, 1)$ generators and single-polylog products, and tabulate those that fail. Carried out at 700 digits, this reports **30** independent new directions through weight 6. The number is wrong. It is an artifact of the spurious scale: the genuine reductions are of large height, and against a basis of a hundred-odd atoms the search cannot see them. (Three searches against bases of 46, 55, and the $(i, 1)$ -plus-single set returned “dimensions” of 3, 6, and 30 respectively—all different, all wrong, the count *decreasing* as the basis grew and the ceiling fell.)

4.2 The shuffle, and the correct count

The shuffle settles it. The key relation is that the shuffle of a (-1) -polylogarithm with an i -polylogarithm produces (i, i) and $(-1, i)$ doubles, for instance

$$\text{Li}_2(-1) \text{Li}_2(i) = \text{Li}_{2,2}(i, i) + 2 \text{Li}_{3,1}(i, i) + \overline{\text{Li}_{2,2}(-1, i)} + 2 \overline{\text{Li}_{3,1}(-1, i)} \quad (10)$$

(verified to $\sim 10^{-694}$), whereas the *stuffle* of the same product produces the $(-1, i)$, $(i, -1)$ doubles. Equating ties all three Gaussian points together. Collecting all 240 stuffle and shuffle relations from the products $\text{Li}_p(s) \text{Li}_q(t)$, $s, t \in \{i, -1, -i\}$, $2 \leq p + q \leq 6$, verifying each to $\sim 10^{-600}$, and computing the exact rational rank of the linear system on the doubles, modulo single polylogarithms and the $(i, 1)$ generators, gives:

weight	unknowns (Re+Im)	dim (stuffle only)	dim (double shuffle)
4	18	8	1
5	24	12	2
6	30	14	2
total		34	5

Theorem 2. *Modulo single polylogarithms and the $(i, 1)$ generators, the Gaussian double points $(i, i), (-1, i), (i, -1)$ contribute exactly 5 new depth-2 constants through weight 6—one at weight 4, two at weight 5, two at weight 6. A representative basis is*

$$\operatorname{Re} \operatorname{Li}_{3,1}(i, -1); \quad \operatorname{Im} \operatorname{Li}_{3,2}(i, -1), \operatorname{Im} \operatorname{Li}_{4,1}(i, -1); \quad \operatorname{Re} \operatorname{Li}_{4,2}(i, -1), \operatorname{Re} \operatorname{Li}_{5,1}(i, -1).$$

Every other Gaussian double of weight ≤ 6 is an explicit combination of these five, the $(i, 1)$ generators, and products of single L -values.

The rank is verified two ways: each relation reproduces its value to 10^{-600} , and the exact-rational rank is precision-independent. As a final gold-check at weight 4, fixing the single survivor $\operatorname{Re} \operatorname{Li}_{3,1}(i, -1)$ to its true value and solving the system predicts all other 17 weight-4 components to $\sim 10^{-692}$.

Remark 3 (the methodological moral). Theorem 2 corrects the integer-relation count of 30 by a factor of six. The reductions are exact consequences of the small-height double-shuffle generators, but the *net* reductions are of large height—so high that an integer-relation search against any adequate basis cannot find them. This is the recurring trap of high-precision experimental mathematics, here biting in the direction of *over*-stating independence. The derived double shuffle, with its exact rank, is immune.

5 The onset of depth three

We now move to depth three and the question that organizes this section: *at what weight does genuine depth-three content first appear in the Gaussian tower?* For the uncoloured multiple zeta values the answer is weight 11. We show that for level 4 it is weight 6.

5.1 The triples live in a two-letter alphabet

Consider the Gaussian triples $t_{a,b,c} = \operatorname{Im} \operatorname{Li}_{a,b,c}(i, 1, 1)$, $a + b + c = 6$. By (4) their Goncharov word is

$$0^{a-1}(-i) 0^{b-1}(-i) 0^{c-1}(-i),$$

since $y_1 = 1/i = -i$ and $y_2 = y_3 = 1/i = -i$: *all three non-zero letters are $-i$* . Thus the $(i, 1, 1)$ triples live in the same two-letter alphabet $\{0, -i\}$ as $\operatorname{Li}_n(i)$ and the $(i, 1)$ doubles. The shuffle (5) of $\operatorname{Li}_p(i) \operatorname{Li}_{q,r}(i, 1)$ therefore lands entirely in $(i, 1, 1)$ triples, expressing a *product*—hence a *depth- ≤ 2* object—as a sum of triples. No regularization is needed: every word ends in a non-zero letter and so is convergent.

5.2 Seven of ten triples are depth two

Taking imaginary parts of the 13 such shuffle products (and the triple products $\operatorname{Li}_p(i) \operatorname{Li}_q(i) \operatorname{Li}_r(i)$) gives 13 linear relations among the 10 weight-6 triples $t_{a,b,c}$, each numerically verified and with exact rational rank 7.

Proposition 4. *Seven of the ten weight-6 Gaussian triples $\operatorname{Li}_{a,b,c}(i, 1, 1)$ are depth- ≤ 2 products of lower $(i, 1, \dots)$ polylogarithms. The triple space has dimension exactly 3 modulo products; the shuffle-irreducible survivors are $(3, 1, 2)$, $(3, 2, 1)$, $(4, 1, 1)$.*

These three are precisely the three Lyndon words of length 6 with three $-i$'s—the free shuffle-algebra generators of the $\{0, -i\}$ tower. The shuffle alone cannot decide whether they are *genuinely* depth three or reduce further to depth-2 constants: such a reduction would be a *stuffle* relation invisible to the shuffle (exactly as $\zeta(2, 1) = \zeta(3)$ drops a word-depth-2 object to depth one). A precision-limited integer-relation search is also inconclusive here, for the same high-height reason as in Section 4. The clean answer comes from the motivic dimension.

5.3 The motivic dimension: genuine depth three at weight six

Deligne's theorem on the motivic fundamental group of $\mathbb{G}_m \setminus \mu_4$ (the Gaussian case) and Glanois's depth-graded basis give the decisive structural fact:

Theorem 5 (Deligne; Glanois). *For level $N = 4$ (and likewise $N = 3$) the motivic Lie algebra is free with exactly one generator in each weight ≥ 1 . Consequently the Hilbert series of the period algebra is $1/(1 - 2t)$, the dimensions are $d_n = 2^n$, and—because the basis is depth-graded, the depth filtration being split—the depth-graded dimensions are binomial,*

$$\dim \operatorname{gr}_p^D(\text{weight } n, \text{ level } 4) = \binom{n}{p}. \quad (11)$$

At weight 6 the dimensions (11) are $\binom{6}{p} = 1, 6, 15, 20, 15, 6, 1$ (summing to $2^6 = 64$), so in particular

$$\dim \operatorname{gr}_3^D(\text{weight } 6, \text{ level } 4) = \binom{6}{3} = 20 > 0. \quad (12)$$

Corollary 6. *Genuine depth-three content appears at weight 6 in the Gaussian tower. The weight-6 depth-three graded space is 20-dimensional; the $(i, 1, 1)$ family's contribution to it is exactly the three shuffle-irreducible survivors of Proposition 4, which are therefore genuine depth three. The full weight-6 picture is: 7 of the 10 $(i, 1, 1)$ triples collapse to depth- ≤ 2 products, and 3 are genuine depth three.*

This is five weights earlier than the classical $N = 1$ threshold, and it resolves a question this program had earlier left open—and indeed had at one point misjudged in the opposite direction, on the strength of a precision-limited search. The same $d_n = 2^n$, “one generator per weight” structure holds for the Eisenstein level $N = 3$, so genuine depth three likewise appears at weight 6 there.

6 The Eisenstein analogue

Everything above has a level-3 counterpart at the cube roots of unity. Write $\rho = e^{2\pi i/3}$, so $\rho^2 = \bar{\rho}$.

6.1 Family 1, with an arithmetic twist

The stuffle closed forms read

$$\operatorname{Li}_{a,b}(\rho, \rho) + \operatorname{Li}_{b,a}(\rho, \rho) = \operatorname{Li}_a(\rho) \operatorname{Li}_b(\rho) - \operatorname{Li}_{a+b}(\rho^2), \quad (13)$$

$$\operatorname{Li}_{a,b}(\rho^2, \rho) + \operatorname{Li}_{b,a}(\rho, \rho^2) = \operatorname{Li}_a(\rho^2) \operatorname{Li}_b(\rho) - \zeta(a + b), \quad (14)$$

with the diagonal corollary $\operatorname{Li}_{a,a}(\rho, \rho) = \frac{1}{2}(\operatorname{Li}_a(\rho)^2 - \operatorname{Li}_{2a}(\rho^2))$. The arithmetic twist is in (14): because $\rho^2 \cdot \rho = \rho^3 = 1$, the colored tail degenerates to the *ordinary* zeta value $\zeta(a + b)$, not a level-3

value. The single cube-root L -values split cleanly under the distribution relation $\text{Li}_n(\rho) + \text{Li}_n(\rho^2) = (3^{1-n} - 1)\zeta(n)$:

$$\text{Re Li}_n(\rho) = \frac{1}{2}(3^{1-n} - 1)\zeta(n), \quad \text{Im Li}_n(\rho) = \text{Cl}_n(2\pi/3), \quad (15)$$

the real part a rational multiple of $\zeta(n)$ (or π^n at even n), the imaginary part the Clausen value—the level-3 analogue of $\beta(n)$, with $\text{Cl}_2(2\pi/3)$ Gieseking’s constant. All of (13)–(15) are confirmed in Wolfram 15.

6.2 The (ρ, ρ) frontier: one constant

Running the level-3 double shuffle as in Section 4—90 stuffle and shuffle relations from products $\text{Li}_p(s)\text{Li}_q(t)$, $s, t \in \{\rho, \rho^2\}$, each verified to $\sim 10^{-600}$, exact rational rank—collapses the (ρ, ρ) doubles still more sharply than the Gaussian:

weight	unknowns (Re+Im)	dim (stuffle only)	dim (double shuffle)
2	2	0	0
3	4	2	0
4	6	2	0
5	8	4	1
6	10	4	0
total		12	1

Theorem 7. *Modulo single polylogarithms and the $(\rho, 1)$ generators, the entire (ρ, ρ) family through weight 6 reduces to a single new depth-2 constant— $\text{Im Li}_{3,2}(\rho, \rho)$, at weight 5—with weights 2, 3, 4 and 6 collapsing completely.*

A gold-check at weight 5 fixes the survivor and predicts all seven other weight-5 components to $\sim 10^{-696}$. The pattern mirrors the Gaussian: there too the weight-5 survivors are imaginary $\text{Li}_{3,2}/\text{Li}_{4,1}$ directions.

Remark 8 (a genuine cube-root obstacle). The decisive shuffle $\text{Li}_p(\rho^2)\text{Li}_q(\rho) \rightarrow (\rho, \rho) + (\rho^2, \rho^2)$ lands only on points with $w = z_1 z_2 \neq 1$, so the computation never needs the mixed points (ρ^2, ρ) , (ρ, ρ^2) , where $w = \rho^3 = 1$. This is fortunate, because those points present a real numerical obstacle: both the Lerch-plus-Wynn evaluator and a direct cumulative-sum evaluator stall there at ~ 8 digits—the period-three, non-oscillating tail defeats Wynn’s ε -algorithm, unlike the even-period $i, -1$ cases, which converge to full precision. Folding the mixed cube-root points in (for the *full* level-3 frontier) requires a period-three-aware accelerated evaluator, or Wolfram; it is the one remaining piece.

7 Methods and verification

Evaluators. Where the built-in `MultiplePolyLog` times out (a second argument on the unit circle), the doubles are evaluated by writing the inner tail as a Lerch transcendent, $\text{Li}_{a,b}(z_1, z_2) = z_1 \sum_{n \geq 1} w^n \Phi(z_1, a, n+1)/n^b$ with $w = z_1 z_2$, computing Φ by a stable forward recursion, and accelerating the unit-circle-oscillating outer sum with Wynn’s ε -algorithm. The triple evaluator carries running cumulative inner sums and accelerates the oscillating outer sum the same way; it is validated against the stored triple values to $\sim 10^{-380}$. *Caveat:* the acceleration assumes a genuinely oscillating outer sum ($w \neq 1$ in the double case; an oscillating leading argument in general). At $w = 1$ it returns a wrong value, and at the cube-root $w = 1$ points the period-three tail stalls every Wynn-type acceleration we tried (Remark in Section 6); such values were taken from Wolfram instead.

Relations and rank. Integer-relation search uses a FLINT LLL over the standard relation lattice, guarded by a Champernowne-number canary (a genuine relation cannot involve the structureless canary, so a non-zero canary coefficient marks a precision-floor artifact) and a smoothness test on the coefficient vector. The double-shuffle computations, by contrast, never search: they generate the stuffle and shuffle relations combinatorially, verify each numerically, and take an exact rational rank.

Two-engine discipline. Every displayed identity was confirmed both in `mpmath` (several hundred digits) and in Wolfram 15, the latter using `MultiplePolyLog`, `PolyLog`, `MultipleZeta`, and `GeneralizedPolyLog`. Two incidental hazards are worth recording: `mpl_wynn` is invalid at $w = 1$ (above); and a multi-line assignment in the Wolfram front end terminates at the first syntactically complete line, so a long relation must be kept on one line or broken only after a binary operator—a truncated relation otherwise reads as a false negative.

8 Summary

For the Gaussian and Eisenstein multiple polylogarithms we have given (i) a complete family of exact stuffle closed forms for the symmetric and diagonal doubles, with the Eisenstein mixed point degenerating onto an ordinary zeta value; (ii) the exact double-shuffle dimensions of the colored depth-2 spaces—5 new Gaussian constants and 1 new Eisenstein constant through weight 6, correcting a six-fold over-count by integer-relation search; and (iii) a resolution of the depth question, showing via a shuffle calculation and the motivic dimension that genuine depth three appears already at weight 6 in both towers, five weights below the classical threshold. The through-line is methodological as much as arithmetic: the derived double shuffle, with exact rank, sees reductions that high-precision relation-finding cannot, and the motivic dimension certifies what neither can prove numerically.

Acknowledgement of tools. The computations use Wolfram Language 15 (and its `MultiplePolyLog` family), `mpmath`, and `python-flint`; the motivic input is Deligne’s theorem on $\mathbb{G}_m \setminus \mu_N$ and C. Glanois’s depth-graded basis for the cyclotomic multiple zeta values.